



**Explosive
Cocktail**

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BY GRAHAM EDWARDS

In *Spectre*, British Secret Service agent James Bond (Daniel Craig) pursues a trail of clues that eventually leads him to the heart of a global terrorist organization headed by Franz Oberhauser (Christoph Waltz). Ably assisted by colleagues 'M' (Ralph Fiennes), 'Q' (Ben Whishaw), and Eve Moneypenny (Naomie Harris), as well as new companion Dr. Madeleine Swann (Léa Seydoux), Bond crosses three continents on his way to an explosive finale in which he confronts not only the dastardly Oberhauser, but also 'C' (Andrew Scott), treacherous chief of the new Joint Intelligence Service.

Emulating Bond's reliance on trusted allies for support, producers Barbara Broccoli and Michael Wilson gathered round them a closely integrated team of familiar faces to help bring the 24th '007' film to the screen. Having cut his teeth on *Skyfall*, director Sam Mendes returned to helm his second Bond movie, supported by special effects supervisor Chris Corbould and visual effects supervisor Steve Begg. Dennis Gassner reprised his role as production designer, while newcomer Hoyte van Hoytema took up the reins as director of photography.

Eschewing the crisp, digital aesthetic employed by his predecessor Roger Deakins,



van Hoytema opted to shoot *Spectre* on 35mm film using anamorphic lenses. “Hoyte was after a very cinematic, romantic look,” said Steve Begg. “That had to be reflected in the visual effects. We had a few tracking and resolution issues — anamorphic is inherently just a little bit softer on the edges, particularly on film — but nothing really profound.” Begg shadowed the main unit throughout an extensive shoot at locations in Mexico, Italy, Morocco, Austria and England, and on soundstages at Pinewood Studios in the U.K. “My main task was to stick with Sam and Hoyte,” Begg commented. “I think it paid off, particularly in post. We ended up talking in shorthand because I had been there when the shots were being made, and I knew exactly what they wanted.”

Under the guidance of Begg and visual effects producer Leslie Lerman, supported by associate visual effects supervisor Andrew Whitehurst, *Spectre*’s 1,540 visual effects shots were divided more or less equally between Cinesite, Double Negative, Industrial Light & Magic, MPC and Peerless Camera Company, with a small amount of additional work overseen by Stuart Bullen at BlueBolt. “The film broke nicely into segments, so we chose the work appropriate for each company,” said Begg. “There was very little overlap between them.”

According to tradition, the main titles of a Bond movie are mini-productions in their own right, filled with surreal moving images set to suitably moody music. The *Spectre* title sequence, the seventh with director Daniel Kleinman at the helm, is no exception. Kleinman worked with long-time collaborator Framestore, reuniting the team behind previous 007 titles including *Casino Royale* and *Skyfall*. “Danny came up with the initial ideas,” said Framestore creative director William Bartlett. “Then we talked about the



themes in the film, and the kinds of metaphors we could use. For example, we knew there was going to be an octopus, and a surveillance theme, so we suggested, 'What about having an eye, only the iris is made of tentacles?'"

The octopus — symbol of the terrorist organization Spectre — became the sequence's dominant image. Early concept development focused on the creature's appearance. "The more you look at octopuses," said Bartlett, "the more you are blown away by just how weird they are. Ours is partly based on a living creature, only stylized to get the look we wanted." Skin color and texture were primary concerns. "We started with something very natural-looking, but it wasn't sinister enough. So we went with a dark, slimy look more in keeping with the criminal organization vibe."

Preliminary lighting and animation tests presented the octopus in an underwater environment. A more menacing creature emerged as the watery cues were sidelined. Throughout, animating the tentacles remained a challenge. "I have never worked on a job where the rigging was going on so late," remarked Bartlett. "An octopus tentacle does not exactly move from the tip, nor does it exactly move from the base. It stretches at certain points and not at others. It was very tricky to rig."

An early version of the sequence featured dancers sporting octopus skin textures. "It looked very beautiful," Bartlett commented, "Octopuses have this incredible texture; but when you see it on a human being, I am not sure you would recognize what it is." Ultimately, cephalopod skin gave way to flames, with practical fire effects composited into a bluescreen shoot of Daniel Craig and his Bond girl entourage. Particle effects added



drifting embers and individual flames trailing from the dancers' hands.

SPECTRE

Octopus and dancing girls finally unite as James Bond plunges into an ink-filled abyss. "We shot the ink in our cloud tank," Bartlett revealed, "We squirted in various types of paint, and used those elements to make the environment. We generated specific bits of the girls' hair using procedural fluid dynamics in Houdini."

Attempting to evade the ubiquitous octopus, Bond experiences visions of former friends and enemies. The 'broken glass' look of these shattered memories repurposed live-action from previous films. "Chaos Group gave us beta test licenses for their new V-Ray plug-in for Nuke," Bartlett said. "That enabled us to do proper refractions and reflections within a 2D system, which handled the live-action more efficiently than if we had done it in 3D." Additional treatment aided the blending of footage shot in multiple formats. "Some of the live-action was on film," Bartlett noted, "and some was on digital. Some was shot underwater. It was all lit differently, so we found a stylized look to bind it all together."

As the sequence concludes, the iconic Spectre ring fills the screen. The Framestore team were keen not to draw parallels with another memorable piece of movie jewelry. "We had a version of the shot where we traveled through the middle of the ring," recalled Bartlett. "But it seemed a bit too *Lord of the Rings*. So we ditched it."

Spectre opens with an attention-grabbing 'oner' in which the camera follows James Bond and his companion Estrella (Stephanie Sigman) through crowds of revelers celebrating Mexico City's Day of the Dead festival. In an apparently unbroken take, they enter a hotel and take an elevator to a room on an upper floor. Bond exits the window, crosses

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the rooftop and targets a sniper rifle on the participants of a secret meeting being held in an adjacent building. Lasting some four and a half minutes, the bravura shot was photographed with spherical lenses to afford more flexibility in post, and combines multiple live-action plates stitched seamlessly together by a team at ILM's London office.

IO Entertainment spent six months developing previs for the opening shot, with available camera movements dictated by an early decision to shoot the opening and closing segments using 50-foot SuperTechno cranes. "We managed the creative choreography and composition of the shot," said previs supervisor Brad Blackburn of IO Entertainment. "We also determined logistics, right down to how long the camera track needed to be, whether we needed to build scaffolding for the grips, and which lamp posts would have to be removed on location so the crane arm could swing through its arc."

The shot begins as Bond and Estrella make their way down a busy street. ILM made numerous fixes to the original location plate, tracking carnival masks to the faces of extras caught looking at the camera, and adding digital crowd extensions. As the couple turns into an arched entrance, the camera lingers briefly on a poster to disguise a soft transition to the second plate, which was shot inside a hotel elsewhere in the city. This and other transitions were implemented by reprojecting the live-action onto geometry, which was then warped to get as close a match as possible between the end of one plate and the start of the next. The performances of the actors dictated the exact timing and position of the blends, with the majority of the warping being applied to the scenery around them.



The second transition occurs as Bond and Estrella enter the hotel room, which was built as a set at Pinewood, with bluescreens placed behind the windows to allow the ILM team to composite in exterior plates of Mexico City. A re-time of the footage created the illusion of a whip-pan as Bond moves to the window. A third transition occurs as he exits through it. Footage of Daniel Craig walking along the roof was captured in Mexico as a single plate, with ILM performing rig removal, adding a gap over which Bond jumps, and compiling replacement backgrounds from multiple sources.

Reaching the end of the roof, Bond takes up a sniping position with his rifle aimed at the window of an ornate building dubbed the 'Fru Fru,' which ILM developed as a digital matte painting based on the building that was shot in Mexico. "The Fru Fru was a bit of a hodge-podge of different bits and pieces," said ILM London visual effects supervisor Mark Bakowski. "We used lidar and total station data from the location for the base model, but we reshaped it and topped it up with a couple more floors. The room behind the window was a studio set filmed separately at Pinewood."

In its final form, the opening shot ends as Bond draws a bead on his target. However, the original intention was to continue the one-take conceit throughout the subsequent action, during which an unexpected explosion causes the Fru Fru to collapse. At one point it was proposed that this — along with the later demolition of the MI6 headquarters — be achieved using large-scale miniatures. "I come from a miniatures background, so I always look for ways to use them," said Steve Begg. "They were very successful on *Skyfall*, and Sam kept pushing for their use. But, when we found out about this elaborate camera move, I immediately thought, 'Forget that — it's going to be CG.'" The crew con-



ducted tests, but ultimately production used no miniatures in the film.

Bond's gunfire triggers an explosion, which causes the Fru Fru's façade to fall forward and demolish the rooftop where he is crouched. The complex stunt was shot full-scale on the Pinewood backlot, with the chain reaction event recorded by 46 cameras. Previs was used to determine not only camera angles, but also the basic geometry of the physical rigs that would be required. "Once the previs was locked down," said Brad Blackbourn, "we looked at it with Chris Corbould to determine what pieces would actually require building, and to define the approximate pivot points of the breakaway sections of the roof."

A special effects team led by Corbould first constructed the front of the building in its final, exploded state, then reinstated the façade with intact masonry made from soft foam, ready to be blown out by pyrotechnic charges. Special effects then built the roof onto which the resulting rubble falls in three breakaway pieces. "We had a section approximately 30 feet by 12 feet that came down a track and impacted with the roof," explained Corbould. "That took out the end third. The next third was triggered to release once that first breakaway piece had gone. The last third, which Bond runs up, hinged down on hydraulic rams. It was all very closely timed, one event after another — and it took months and months of planning and testing." Although the entire rig was capable of being reset, the complete event was captured successfully on the first take. Multiple takes on just the final hinged section allowed the stunt team to perfect Bond's scramble over the disintegrating roof.



Using the resulting live-action as a template, the explosion and collapse were re-created in digital form. “Chris’ rigs helped us work up a believable camera move, with the stuntman running over the collapsing roof,” said Steve Begg. “Once Sam and the editors were happy, we gave those elements to ILM, who pretty much replaced everything.”

ILM’s digital building was constructed in Autodesk Maya and textured using The Foundry MARI and Adobe Photoshop. “We built it up using individual components, then applied material-based destruction to them,” Mark Bakowski explained. “At the base of everything was keyframe animation, with the building pre-modeled to bend and buckle appropriately. Once we had the basic collapse blocked in, we dressed in multiple secondary simulations for dust, rebar, rubble, glass, and so on. We could re-block it all very quickly — it was very light on its feet.” Effects were integrated in Side Effects Houdini, with final renders output through Pixar’s RenderMan.

ILM also married a digi-double head of Daniel Craig to his stunt double’s torso, which ended up being the only part of the live-action retained from the plate. “It was an interesting challenge keeping the CG head looking like Daniel Craig,” Bakowski observed. “The minute we put too much of a performance into him, he started to look particularly un-Daniel-Craig-like! We slightly over-emphasized certain features, like his chiseled cheekbones. We would keylight those a little bit, slightly contrary to shot lighting, to give him that classic Daniel Craig look.”

Bond’s primary target, international assassin Marco Sciarra (Alessandro Cremona), survives the explosion and flees into the crowded Zócalo square for extraction by helicopter. Around 1,500 extras were present during principal photography, but for many



shots ILM swelled their numbers considerably. "A crowd of 1,500 is fairly substantial when the camera is low," commented Steve Begg, "but when you climb above six feet you start to see where it runs out. Hoyte kindly agreed to switch off his smoke machines, which would have caused us problems putting in the crowd extensions on all but the lower shots. That helped immensely."

Working primarily in Massive, ILM crowd supervisor Jo Plaete choreographed up to 50,000 digital extras using a movement library based on a one-day motion capture session at Shepperton Studios. "At the start, it's a gentle parade," observed Mark Bakowski. "As time goes on, it develops sort of a concert atmosphere. Then the people scatter and panic when the helicopter arrives." Individual crowd members were fashioned using photogrammetry of extras gathered by Propshop, who facilitated asset creation throughout the production by scanning locations, sets, vehicles and actors. While ILM's mandate was to replicate the Day of the Dead costumes exactly, the overall color palette was expanded somewhat for wide shots. "Lots of people were wearing black and white. So once we pulled out to show a few thousand people, it started to get very monochrome. We tried to add variety while keeping it looking realistic."

As the helicopter lifts Sciarra to safety, Bond leaps aboard and a brutal fight ensues, during which the chopper performs a series of violent maneuvers over the teeming Zócalo. "We choreographed four or five minutes of continuous choreography as a single, massive sequence," recalled Brad Blackburn. "Then we started putting in camera coverage. Once Sam was happy creatively with the sequence, we were able to work out what



kind of movement the camera helicopter would have to do, and exactly pinpoint every camera position that was needed in the square or on the rooftops.”

During the shoot, stunt pilot Chuck Aaron took the MBB Bo 105 helicopter through its paces over a largely empty Zócalo. More extreme aerobatics were enacted over an airfield south of Mexico City. “Sam was initially quite negative about using CG,” Steve Begg remarked. “Then, a few weeks before we were due to shoot in the Zócalo, we were denied permission to do the more serious stunt work over the square — barrel rolls and things like that. So I hit Sam with Plan B, which was to use the runway lines on the airfield as tracking markers for a CG Zócalo. And he was cool about that.”

Using photogrammetry gathered by a drone pass of central Mexico City, as well as hero lidar and texture passes, ILM created a fully digital environment that faithfully reproduced the Zócalo, plus two layers of buildings beyond it. Streets outside this zone were laid out using a generic library based on local architecture. The distant skyline employed 360-degree photography of the city projected onto a planosphere. “It was one large, multi-purpose asset, which we matte-painted up for each specific angle,” explained Mark Bakowski. “We used a lot of dressing — satellite dishes, water tanks, aerials, flags — to sell the sense of scale and parallax when the helicopter was flying over.”

Shots featuring the digital Zócalo also employed a CG helicopter. In replacing the MBB Bo 105 with its digital twin, ILM took pains to match the appearance and movement of the real aircraft. Motion-captured digi-doubles were used for the figures fighting inside and outside the cabin. “Replacing the helicopter saved us the worry of extracting it from the background, particularly with the rotor blades and all the motion blur,” Bakowski



stated. "We tracked everything to get the motion just right, so no one could say, 'Hey, that's not a realistic helicopter move.'"

For closeup shots, real helicopter body shells were strengthened and mounted on three different rigs in front of bluescreens at Pinewood. The first was a conventional hydraulic gimbal with up to 35 degrees of movement in each of its six axes. Because the script called for the helicopter to perform loops and barrel rolls, Chris Corbould also devised 'spit-roast' rigs. "We mounted the helicopter cockpit with pivots in the nose and tail, so we could spin it along its length," Corbould said. "Then we adapted that rig so we could somersault it end over end." To create a safe environment for the performers, Propshop fitted out the helicopter shells with interiors made from polyurethane rubber, and also replaced certain external parts with the same soft material. Multiple witness cameras were used in addition to conventional bluescreen markers, to maximize the tracking data available from the rapidly-moving rigs.

ILM composited its digital Zócalo environment into the backgrounds of the bluescreen shots, extended the bucks where necessary, and added whirling rotor blades and reflective window glass. "Because the bluescreen helicopters had various sections made of rubber, they distorted slightly when the stuntmen or actors were pulling on the edges," stated Mark Bakowski. "We had a few problems inserting our glass into the warping edges of the windows."

Bond dispatches Sciarra and returns to London, only to find himself suspended from duty. Defying his curfew, he travels to Rome and gains access to a secret gathering of a shadowy organization led by the mysterious Franz Oberhauser. Exteriors for Ober-



hauser's palazzo were shot in the U.K. at Blenheim Palace. Cinesite created a 3D model of the palace, replacing the baroque roof with simpler Romanesque architecture. The redesigned roof was then composited into the live-action using The Foundry Nuke. For a high-angle establishing shot, the complete CG building was nestled into a plate shot over the tree-lined Via Garibaldi in Rome. "It was a tricky shot, with layers and layers of parallaxing tree branches as the camera pans up over the top," commented Cinesite visual effects supervisor Zave Jackson. "We managed to get a decent matte, then we put in 2½D grounds surrounding the palazzo, and blended that into the existing environment."

During the secret meeting, Oberhauser's newest henchman Mr. Hinx (Dave Bautista) gouges out the eyes of a rival assassin. Cinesite tweaked the footage to compensate for the understandable caution exercised by the actors during the shoot. "Dave Bautista was concerned that he was going to hurt the other actor," said Jackson. "His thumbs were jittery and not really in the right place. We digitally manipulated his thumbs to force them much closer into the eye sockets, and added a bit of blood."

When Oberhauser spots Bond, the British agent flees in his recently-acquired Aston Martin DB10, only to be chased through the Italian capital by Mr. Hinx, driving a Jaguar C-X75. The DB10 was developed specifically for *Spectre* by a design team working under Aston Martin's chief creative officer Marek Reichman. "We worked very closely with Aston Martin and Jaguar technicians," said Chris Corbould, who supplemented his special effects duties by heading up *Spectre*'s action vehicles department. "We tested those cars almost to destruction. Our first priority was obviously the stunt drivers and any people



who were going to be in the vicinity, but in Rome we also had 2,000-year-old buildings to look after. By the time we came to shoot the chase, the stunt drivers knew every inch of those cars and knew exactly how they were going to perform.”

Eight hand-built Aston Martins were used in total, four of which were adapted for stunt work. Two others had control pods fitted on the roof, enabling the actors to perform inside while stunt drivers operated the vehicles from above. The remaining two cars were kept pristine. Propshop, working closely with Dennis Gassner, supervising art director Chris Lowe, and set director Anna Pinnock, assisted in transforming the Aston Martins from concept cars into fully functioning action props. “The DB10 had a very basic interior,” observed Propshop’s asset producer James Enright. “We modified the steering wheel and dashboard, and added the switches that operate all the gadgets. It was important the car didn’t look overproduced. It was straight from Q’s workshop — that’s why the interior is a bit rough and ready.”

Cinesite enhanced bullet hits in the scene in which Bond drives away from Oberhauser’s palazzo. “Sam Mendes had a very particular idea of what he wanted the sparks to look like,” Zave Jackson recalled. “He didn’t want them to look like classic zirconium pellet hits, because that’s what everyone does. We did a lot of manipulation of the practical elements that had been shot, adding bits of green flare to give them a slightly colder look.”

The Cinesite team also enhanced wide shots of the city at night, incorporating 2½D digital matte paintings developed from bracketed photography of the real locations, and relocating iconic Roman landmarks to create aesthetically pleasing compositions. Environmental fixes included painting out the many production lights used to show off the



supercars to their best advantage. "That's the nature of what the Bond work is," Jackson observed. "It's invisible effects playing a supporting role rather than a starring role."

Stunt coordinator Gary Powell supervised the high-speed chase, during which both vehicles frequently topped 100mph. For a shot in which the Aston Martin flies through the air and takes the roof off a parked Alfa Romeo, a cable was used to detach a breakaway section of the stationary car's bodywork a split-second before impact. "Sam was keen for the roof to fly up in the air," Chris Corbould said. "But we found that if you just drove into it, the roof wrapped around the Aston Martin and looked messy." Cinesite removed the cable and added a spray of digital glass erupting from the empty windshield of the Alfa.

Two of the stunt vehicles were fitted with a rear-facing flamethrower, which Bond deploys as Mr. Hinx pursues him along the banks of the River Tiber. "We had a 40-foot flame coming out of the back of the Aston Martin and engulfing the Jaguar," said Corbould. "We used isopropanol alcohol as opposed to gasoline because we didn't want to get pollution in the water. We also had to make sure that the Jaguar's air inputs were kept clear."

The chase ends when Bond ejects, leaving the now-driverless DB10 to plunge into the river. On location beside Rome's historic Ponte Sisto, a pneumatic ram was used to accelerate the car up a shallow ramp and launch it into the water. A separate tether cable ensured the Aston Martin would never get closer than 30 meters to the 15th-century bridge. "We did tests at Longcross Studios to make sure we knew exactly where it was



going to land,” Corbould stated. “We added buoyancy devices to the car that were activated as soon as it hit the water. The current swung it in towards the bank and we pulled it out with a crane.”

Car interiors were shot on stage at Pinewood in front of rear-projection screens. Crews gathered projection footage in Rome by a vehicle, which was equipped with a nine-camera roof rig, driving the complete route of the chase. The resulting 70 minutes of multi-angle footage were then stabilized by Cinesite. Matched pairs of projectors were used on-set, with identical plates overlapped in perfect alignment so as to maximize the brightness of the image. “The benefit was that we got nice interactive lighting on the actors sitting in the car and across the dashboard of the car, and interesting reflections in the windows,” Zave Jackson remarked. Around half the rear-projection shots benefited from additional work by Cinesite. “We reinserted the same footage that had been projected, only we had the flexibility to intensify the lighting levels, adjust the focus, tweak the horizon line and add camera shake. When the actors waggled the steering wheel, we could animate the background slightly to give some justification to that movement.”

While Bond continues his mission, M’s new rival, C, presents a controversial surveillance initiative called ‘Nine Eyes’ to a delegation in Tokyo. The sequence was shot at City Hall, London, with night views of the capital replaced by a digital matte painting created by Eglantine Boinet at Peerless. “It was shot without greenscreens,” said Peerless visual effects supervisor Paul Round, “so there was a lot of tricky roto, done by Jarek Ancuta and Elliott Round. Take Andrew Scott as C — black hair against a black skyline! Also, there was a glass balustrade behind the windows, so it had to look like you were



seeing the city through two layers of glass. We added atmospheric noise and bokeh light effects — all those little distortions that make it look alive. Those shots were comped in Flame by Paul Gill.”

Traveling to Austria, Bond seeks the help of psychologist Dr. Madeleine Swann. Exteriors of Dr. Swann’s alpine clinic were shot at the Ice Q restaurant on Austria’s Gaislachkogel Peak. Working from production art, artists at MPC Vancouver used Autodesk Maya to create a digital model of the clinic, which replicated the existing Ice Q architecture four times around a central courtyard. For establishing aerial views, digital matte painting lead Daniel Villalba-Sunol perched the model atop 3D mountain geometry, onto which reference footage of the location was projected.

MPC also enhanced location shots of Bond approaching the clinic on foot. “The left side of the real building was in the plate,” recalled MPC visual effects supervisor Mark Curtis, “but the glass was reflecting a flat, gray sky where what we really wanted to see was sunlit clouds and mountains. We used our model to replace the whole building apart from the entrance and walkway, then rendered reflections back into it. We included the interior structure that gets refracted and reflected through all the different layers of glass, and added blowing snow to give it a little texture and movement.” Production shot clinic interiors on a Pinewood soundstage, with views through the panoramic windows captured in camera using location photography on a wraparound Translite backdrop measuring some 50 by 500 feet. MPC did minor cosmetic work to add subtle cloud movement and snow effects to the otherwise motionless alpine vistas.



A squad led by Mr. Hinx abducts Swann and speeds down the mountain in off-road vehicles, while Bond gives chase in a Britten-Norman Islander aircraft. As with the helicopter, closeups of the plane were shot with a buck mounted on a hydraulic gimbal. “Unfortunately for us, they chose a highly reflective, shiny black plane,” said Steve Begg. “We used a white cyclorama to give us this lovely alpine light on the interior, but the exterior was CG.” MPC created a number of Islander models in varying stages of destruction, including one riddled with bullet-holes, and switched between them as the sequence progressed.

To generate reflection maps, MPC flew a virtual camera down a digital replica of the Austrian location. “Sometimes we put our reflections directly onto the live-action,” Mark Curtis commented. “Other times it was more expedient — and looked better — to replace the whole plane.” A special tech-anim rig was developed to replicate the vibration effects generated by the gimbal.

The chase continues through an avenue of snow-covered trees, which eventually narrows to the point where the plane’s wings are sheared off. “There is fairly extensive effects work there,” observed Steve Begg, “a marriage of practical and digital.” Live-action for the sequence was shot at an Austrian location where, conveniently, an overhead power line had been routed through a stand of trees. The local authorities agreed to the temporary removal of the pylons, allowing Chris Corbould’s team to string a cable between two cranes positioned approximately 1,500 feet apart at opposite ends of the resulting tree-lined corridor. A glider winch was used to propel a real aircraft — suitably lightened and strengthened — along the cable at speeds averaging 50 miles per hour.



In one shot, the wheel of the Islander impacts the roof of a fleeing vehicle. “We adjusted the crane so we knew the plane would just clip the vehicle,” Chris Corbould recalled. “But then, weirdly, when we actually did it at full speed it ended up five or six feet above the height we had set it at. In the end, we deduced we were pulling the plane so fast it was actually flying. We had to slow it down to stop it trying to take off! We were quite proud of that.”

For successive shots in which the Islander is seen descending below tree height, MPC effected a seamless transition between the wire-rigged plane and their CG model. They also tweaked backgrounds to bring the trees closer. “Sam felt there wasn’t enough danger around the plane,” said Mark Curtis, “so we added CG pine trees, with a digital matte painting of a wall of trees behind, and banks of snow pressing inwards.”

Breakaway wingtips detached at the moment of impact when the aircraft smashes into the trees. “The last ten feet of the wings were soft and lightweight,” Chris Corbould said. “We lined the plane up with a real tree on one side, and a tree we had planted on the other. Then we literally flew it into them and released the wings as it hit.”

Bond steers the grounded aircraft down a slope and into a large timber shed. Bursting through the back of the building, the plane collides with Mr. Hinx’s vehicles, bringing the pursuit to a dramatic halt.

Two wingless versions of the plane were built around a pair of double-tracked Ski-Doo snowmobiles. Bespoke nose skids allowed the crippled Islander to be maneuvered with relative ease through the snow. Nitrogen cannons severed the tail of the plane as it hit the ground, and launched the entire craft through the back of the shed — a structure



built on location by the construction art department. Cables guided the plane to its final point of impact with the stunt vehicles.

The visible nose skid of the Ski-Doo plane was removed by MPC, which also added spinning propellers to the engines. Effects lead James Burke simulated snow plumes — using Flowline, Maya Fluid Effects and in-house tools PAPI and Kali — that were rendered in Pixar's RenderMan RIS. "We broke the snow up into three levels," Mark Curtis explained. "Underneath was a dense, packed layer. Chunks of that got ripped up by the nose and thrown backwards. Then there was a lighter layer of fresh snow that was churned up by those chunks. We also added little streamers of powdered snow coming off the chunks. The third level had all of that getting a blast of air from the engines. In some cases we ended up rendering 20 or 30 different layers."

Replicating the appearance of the propellers posed its own subtle challenges. "Primarily, they were modeled, textured, rendered propellers," remarked Curtis, "but you couldn't just look at them static, think, 'Yeah, that looks right,' and then spin them. We ended up with an additional 2D setup where our compositing supervisor, Jan Dubberke, used cards to add a fluttering texture, or to diffuse or darken the propellers in places against the background." MPC also sweetened the shot of the plane bursting through the shed wall by adding effects animation of sawdust and wood splinters as the propellers chewed through a pile of logs.

Revealing the name of Oberhauser's organization — Spectre — Swann leads Bond to L'Americaïn, a hotel in Tangier, Morocco. For establishing shots, Cinesite replaced the unprepossessing view beyond the Grand Socco market with an expansive digital matte



painting of the Bay of Tangier, and created a fully CG hotel exterior extrapolated from the interior set built at Pinewood. Another digital matte painting was used to portray the view through the hotel windows, which had been simulated on set by a scenic painting with three-dimensional foreground elements. "The initial thought was for us just to add chimney smoke and birds flying past, but that wasn't enough," noted Zave Jackson. "We replaced it all with a full-blown re-creation of the environment based on reference photographs of the bay. That involved fairly detailed rotoscoping, because of all the little diamond shapes in the Moroccan-style windows."

The amusing antics of a hotel mouse lead Bond to a hidden room where abandoned equipment helps him pinpoint the location of Spectre's secret base. Cinesite CG supervisor Chris Petts oversaw the creation of a fully digital rodent, which was composited into plates from the Pinewood shoot. "Sam sent us a couple of photographs of what he was looking for," Jackson recalled. "He wanted the mouse to have a matted, scruffy look to its fur, and we put a lot of effort into grooming our mouse to get the texture he was after. Sam also gave us quite specific feedback on its movement." The mouse was modeled in Autodesk Maya and rendered with Solid Angle Arnold, with fur created using Peregrine Labs Yeti.

Seeking the Spectre base, Bond and Swann take a train to a remote location in the Tunisian desert. Scenes of a locomotive hauling four carriages were shot in Morocco, with Cinesite adding three additional passenger cars and lengthening the trailing goods carriage. Starting with a lidar scan of the practical rolling stock, Jackson's team adapted the configuration of the train to accommodate the action demanded by the script and



defined by the layout of the interior sets at Pinewood. "The main action starts off in the restaurant car," Jackson explained, "then moves into the bar carriage, then the kitchen section, and finally the goods carriage. We modified the internal carriage and window details of our CG train accordingly."

The arrival of Mr. Hinx prompts a close-quarters fistfight between Bond and the burly assassin. Head replacements were especially critical for shots in which Mr. Hinx is set on fire. "We created digital head assets for Bond and Hinx to replace the stunt performers' heads in a variety of shots," said Jackson. "Our CG shots were intercut back to back with shots of the real actors, so we were very aware of the need to fit them in seamlessly. A lot of work went into developing the full CG heads, particularly the hair groom and lighting to match the fire-lit environment."

A knee injury suffered by Craig while performing the fight scene caused the shoot to be suspended for several weeks, during which time the previously-connected train carriage sets were separated. When filming resumed, greenscreens were placed at the open ends of each isolated carriage, and Cinesite reinstated the view of the missing carriage beyond. "We did a temporary version using still photographs," said Jackson, "but even though it's cut incredibly fast, it simply didn't work — the camera moves were just too three-dimensional and complex. We ended up reconstructing the interiors with 2½D projections using images that we'd captured before they struck the set."

Mr. Hinx meets his end in the goods carriage, when a rope attached to a string of barrels yanks him off the train. A short section of rope was attached to Dave Bautista's



neck, with red tape wrapped around the ends for tracking reference. Cinesite attached a coil of digital rope to the free ends and animated it to perform the requisite whipcrack motion. The nighttime view of the desert seen through the open carriage door behind Hinx comprised a live-action plate shot at dusk, graded and manipulated by Cinesite. "I was dreading doing that shot with stars and a beautiful glow on the horizon — all that stuff which normally screams out 'fake CG!'" Steve Begg remarked. "Sam let me run with a bit of desert near the train illuminated only by the light from the carriage, with the occasional whiff of dust and a telegraph pole or two. It was a lot more realistic."

Deep in the desert, the Spectre base is revealed as a cluster of buildings nestled inside a meteor crater. For exterior scenes, a minimal set was built adjacent to an extinct volcanic crater near Erfoud, Morocco. Working from production concept art, a team at ILM's Vancouver office, led by visual effects supervisor Greg Kegel, developed the architecture and layout of the base, incorporating set extensions, palm trees and a wide array of additional structures. They also created the complete surrounding terrain as 3D geometry, using textures mapped from photographic surveys of the location. "The real crater was there, just not at the right scale," commented Mark Bakowski. "We used it as a guide, but any time you see the interior of the crater it's all digital."

Following a tour of the Spectre base, Bond is taken to a torture room and strapped to a sinister-looking chair equipped with a fiendish device designed to drill into his skull. "For research, we took a dentist's chair apart," Propshop's James Enright recalled. "Then we scanned Daniel Craig, 3D-printed a dummy of him, and built the final prop around it. This meant we could work on the chair without having to get Daniel off the set every five



seconds.” Propshop created the chair components digitally before fabricating them in metal. “On the day, we had the arms physically there but without the drill bits. The arms were radio-controlled with tiny servos, as were the neck and arm braces, which had to ping open during the escape sequence.”

The practical arms were ultimately filmed in static poses and replaced with digital counterparts by a team at MPC London. “The practical chair had quite a lot of movement in it,” stated MPC visual effects supervisor Jessica Norman, “so the key was getting it to feel like the digital arms were really attached to it. We slightly changed the design to allow the arms to tuck in at the back of the chair. We animated them to get a feeling of precision in their movement, but with a bit of a ‘click’ as they came to rest.” Closeups of the drill bits boring into Bond’s flesh required accurate tracking of the live-action plates. “They drew a little dot on Daniel Craig’s face for where the drill was to go in. Unfortunately, it was very hot on set, so the dot ended up in a different place each time! We roto-animated his face to make the drill stick to one spot, and created a CG skin patch with a little indent around the needle.” To create reflections on the metallic surfaces of the torture device, MPC generated digital versions of the environment, Bond and the other characters in the room. The view of the crater base through the windows was created on set using a painted backdrop, which was replaced by MPC using The Foundry Nuke to composite in sections of a 360-degree digital matte painting provided by ILM.

Shortly after Oberhauser reveals his true identity — Ernst Stavro Blofeld — Bond escapes through the timely use of an exploding wristwatch. Peerless delivered several shots in which the watch’s analog dial transforms into a countdown timer. “There were a lot of



reflections in the watch face,” observed Paul Round, “particularly in one shot where Madeleine Swann is looking at it. You couldn’t really replace all that with CG.” Working in Autodesk Flame, Round isolated the reflections, painted out the original watch face and tracked in 2D animation of the glowing red countdown graphics.

Bond’s detonating watch sets off a colossal explosion that destroys the entire Spectre base. The pyrotechnic event — for which Chris Corbould was subsequently awarded a Guinness World Record for Largest Film Stunt Explosion — was filmed on location and used more than 2,000 gallons of kerosene in conjunction with 33 kilograms of high explosive. Early previs explored the choreography of the event in rough form, with test footage projected onto cards in a 3D environment. In Morocco, Corbould’s team worked with explosives specialists at Event Horizon to design and engineer the record-breaking conflagration, which took two weeks to test and rig, and covered an area equivalent to ten football fields. “It took 25 to 30 different explosions to make up the one big explosion,” Corbould noted, “and there were probably ten different components to each of those. We had about 200 detonators going off, each one programmed to within a thousandth of a second.”

On the day of the shoot, the ‘fire’ instruction was triggered via a laptop command, after which there was a built-in three-second delay to allow all the detonators to synchronize. “That was the scary thing for Sam and Daniel,” said Chris Corbould. “We were literally pressing the button halfway through Daniel’s line. Once we pressed it, there was no way back. If he had got the line wrong, it could have been disastrous.”



Corbould appreciated Sam Mendes' decision to shoot the massive explosion with a single camera. "Sam was quite clever, really," Corbould commented. "It would have been a great temptation to have a dozen cameras on that explosion, and to keep cutting to various angles of cameras being engulfed in flames, and debris flying past the lens. I really liked the way he just kept it to one camera, as a sort of off-the-cuff event going on behind Bond."

The final explosion plays out in a single, long take, which proved challenging for the ILM compositors when the time came to add in the base architecture. "We had to layer in all the different elements of our CG base at different depths within the fireball," Mark Bakowski explained. "But there was no easy way to separate out each fireball from the one behind, because obviously there were no bluescreens in there. A lot of hard work went into the interactive lighting on our buildings to sell that. We added little bits of debris flying off here and there and a lot of smoke to really get the interaction."

Back in London, Bond reunites with M for a showdown with the nefarious C, now revealed as a Spectre stooge. Shot primarily in summer, many of the London scenes required cosmetic work to replace trees in full leaf with their bare-branched equivalents. For a nighttime aerial shot of Trafalgar Square, Peerless added leafless trees generated in Autodesk Softimage to a live-action plate captured by Helicopter Film Services, which gained permission from the Civil Aviation Authority to fly an Intuitive Aerial Aerigon UAV drone through the normally restricted airspace.

Dominating the London sequences throughout the film are two key structures: the former MI6 headquarters, bomb-damaged and slated for demolition, and the ultra-modern



Centre for National Security, within which the ambitious C is building his new intelligence empire. Facing each other from opposite sides of the River Thames, both structures were brought to the screen by Double Negative. "Sam Mendes had very clear ideas on how he wanted the CNS building to appear," said Alex Wuttke, who took over from Jonathan Knight as Double Negative visual effects supervisor late in postproduction. "The building was located in a very specific position, so there could be shots looking out from it towards the dilapidated MI6 building." CG supervisor Joel Green managed the construction of a fully CG CNS building, which was then composited into live-action plates of London, with views of the city reflected into the glass walls using HDRI panoramas captured from the real-world location. "The nighttime shots were the trickiest. Sam didn't want it to look super-futuristic and *Blade Runner*-esque — which is where we were to start with — so we trimmed back the amount and complexity of the reflections to give it a simpler, more graphic silhouette."

Separated from Bond, M finally confronts C inside the CNS. Interiors were shot on a multi-level set at Pinewood, with a large Translite providing views of the city beyond. Double Negative later rotoscoped the foreground elements and replaced the Translite with a digital matte painting, adding life with a digital River Thames, moving vehicles and atmospheric effects. CG set extensions elevated the studio set to a position two-thirds of the way up the building, notably in high-angle shots looking down into the structure's central atrium. "It was a good decision to shoot with the Translite," Wuttke commented. "The whole building is glass and steel, so it meant that all the surfaces reflected what they were supposed to be reflecting."



Bond is captured by Spectre agents and taken to the abandoned MI6 headquarters. Visual effects realized the tumbledown structure using a similar methodology to that employed for the CNS building, with a CG model serving for exterior views, and a digital matte painting delivering the panoramic views of London seen from the blast-damaged upper office. Aware that the script demanded the building's ultimate destruction, Double Negative chose to rebuild it from scratch rather than re-use assets from *Skyfall*. "Whenever you are destroying structures," explained Wuttke, "there are always certain requirements for how you build the geometry in the first place."

Digital extensions expanded the MI6 interior sets, such as in a Steadicam shot of Bond looking up through a piece of netting into an enormous hole perforating the building's many floors. "There is nothing real there apart from Bond," said Steve Begg. "We had a net for lighting purposes, but from there up it's fully CG."

More extensions served to enhance a dock area, with Double Negative adding an exit tunnel, digital water and bobbing boats. They also enhanced shots in which arrows mysteriously drawn on the walls guide Bond through the labyrinthine interior. "The arrows were all added in post," Alex Wuttke stated. "It was a late decision by Sam to make the storyline more cohesive."

In the depths of the MI6 building, Bond once more encounters Blofeld, now disfigured by a hideous facial scar. Working under makeup department head Naomi Donne, prosthetic makeup designer Mark Coulier created the final look of the iconic character, first exploring scar designs — with concept artist Simon Webber — over a lifecast of Christoph Waltz. "Sam is very particular," said Coulier, "so we presented seven different versions



of the makeup. We also composited each version into a photograph of Christoph, and did fully artworked renditions of that. Then we narrowed it down by re-sculpting it from all the bits that Sam liked." Following tests, the makeup was further refined. "I went in on a Sunday, and re-sculpted the whole lower part of the makeup according to all these little notes that Sam had made about the scar positioning and the way it looked. That ended up as the final one we used."

Sam Mendes was keen for Blofeld's appearance to reference the version of the character played by Donald Pleasence in *You Only Live Twice*. "The positioning is similar," Coulter observed. "Sam wanted the deep cut through the eyebrow, so we did a silicone prosthetic over Christoph's brow. David Malinowski punched all the eyebrow hairs in." A separate silicone piece extended the scar from below the eye down the cheek. Makeup artists Josh Weston and Vicky Bancroft took approximately one hour to apply the complete prosthetic makeup to Waltz.

As originally conceived, digital effects would be added to show Blofeld's eye as being completely gouged out. "We had this huge black socket for quite some time," commented Steve Begg. "It was so convincing that it was actually quite hard to look at. Ultimately Sam said, 'This is just too horrific.'"

The final design saw Double Negative adding a cloudy, blinded eyeball within the damaged socket. "As it turned out," Alex Wuttke related, "the eyelines were much easier to read, and Christoph Waltz's performance came through a lot more readily. We had to gather a bunch of rather disturbing reference material of people who had damage to the eye, which was fairly gruesome."



Blofeld departs, leaving Bond to track down the kidnapped Madeleine Swann and escape in a speedboat just as the MI6 building explodes. The initial destruction sequence was shot at Pinewood, using pyrotechnic charges placed throughout the 150-foot-long set and triggered at one-thousandth-of-a-second intervals. "We set it up like a demolition, where everything is interlinked so the whole lot doesn't go off at the same time," Chris Corbould remarked. "If you didn't do that, the shock wave would be enormous, especially on a soundstage." As Bond and Swann flee through the collapsing structure, Double Negative integrated additional layers of dust and falling masonry to heighten the sense of peril.

For wide shots of the destruction, effects supervisor Andrew Cadey systematically demolished Double Negative's CG building. "We started off working with an over-length simulation that covered all the major beats," Alex Wuttke explained, "which we used to get signoff on the various components of the destruction. Then we dropped the simulation into the shots that it was going to inhabit and tweaked timings according to the cut. Once everyone was happy with that, we implemented secondary simulations like concrete dust, small pieces of glass, and pyroclastic dust clouds pouring out of the building." Artists created the simulations in Side Effects Houdini, and used Houdini's built-in renderer, Mantra, for the MI6 collapse to optimize the integration of effects elements with the hard-surface destruction simulations.

Bond steers the boat down the Thames in pursuit of Blofeld, who is fleeing across London in a helicopter. To replace summer foliage lining the river embankment, 2D supervisor Daniel Rauchwerger used digital matte paintings of winter trees. "We also had a library of 3D trees," noted Wuttke. "If the shot was fairly close and had a lot of parallax,



we would switch to fully CG skeletal trees. We mixed up the techniques according to the requirements of the shot.” Traffic lockdowns during the night shoot meant that many of the roads and bridges visible in the live-action plates were empty. Double Negative added CG vehicles to the plates, their movements choreographed to suggest that drivers were slowing down as they witnessed the chase.

A lucky shot from Bond strikes the helicopter’s engine, setting it on fire and precipitating a plunge towards Westminster Bridge. Double Negative added flames and smoke trails, and in certain shots aided continuity by tracking footage of Christoph Waltz into the stricken chopper. The crash itself was a practical effect shot on Pinewood’s 007 stage, on a set that re-created a section of Westminster Bridge in near full scale. “We worked in previs with Chris and Steve to look at different trajectories and impact points,” said Brad Blackburn. “We blocked all that out based on the actual location, with a semi-transparent version of the 007 stage sitting over it so Chris could work out optimum angles for the rig that was going to drop the chopper onto the bridge.” IO Entertainment’s previs allowed Sam Mendes and Hoyte van Hoytema to choose from around 30 different camera angles when constructing the sequence.

Crews deployed a full-scale helicopter on set, suspended from a rig similar to that used for the tube train crash in *Skyfall*. “We had two real helicopters that we gutted and then put steelwork in so that we could hang them from an overhead track,” explained Chris Corbould. “We had arms coming down off the track to two points on the top of the helicopter. We were literally puppeteering it. The track allowed us to winch it in at high



speed, drop it down, skid it across the top of the bridge, and then turn it sideways and crash it into a lamp post at the end.” Although the first take worked perfectly, Corbould was keen to push the envelope: “I said to Sam, ‘Now we’ve got one safe in the can, let’s really crank it up.’ We brought it in at literally twice the speed, and it looked a hundred percent better.” The asphalt surface through which the helicopter plows was built in breakaway sections and laced with pyrotechnics to throw off sparks.

For the shot in which the helicopter impacts the bridge, Double Negative replaced the practical buck with a CG chopper. “The general feeling was that the movement wasn’t quite as convincing as everyone wanted,” said Alex Wuttke. “But, because we had the practical helicopter in the plate, we had really great lighting reference so we could match our CG in seamlessly.” Effects simulations added to the illusion of the helicopter tearing up the road surface, and raising clouds of dust and smoke.

Once again, the Translite surrounding the set was replaced by a 2½D matte-painted environment, with foreground buildings fleshed out in 3D. “They let us shoot some plates from the middle of the bridge of all the assembled police cars and soldiers and various personnel,” said Steve Begg. “So we had these lovely elements with all the flashing blue and red lights — all that really brought it to life.”

With Blofeld in custody, all that remains is for secret agent 007 to drive into the dawn with his latest Bond girl at his side. As he departs, Bond’s route takes him past London’s historic Banqueting House, which on the day of the shoot was covered in scaffolding. Peerless replaced the building with a clean version using photographic projections of the 17th-century architecture. “The scene was shot at four a.m., so it was pretty



cold,” Paul Round recalled. “We put a heater in the car to demist the windows, which created a heat-haze coming off the roof. We had to massage things in comp to replicate that. But it was nice to get the last shot in the movie.”

The vehicle in which Bond makes his exit is one guaranteed to bring a nostalgic smile to the face of every 007 fan — the Aston Martin DB5, which first appeared on screen in *Goldfinger* in 1964. Indeed, with its blend of globetrotting action and arch one-liners — not to mention the reintroduction of Bond’s most infamous archenemy — the 24th James Bond film has edged the whole franchise perceptibly closer to its roots. However, thanks to a director clearly able to wield modern filmmaking tools with intelligence and aplomb, *Spectre* is anything but backward-looking.

“I come from a hugely practical background,” noted Steve Begg, “and I love trying to get stuff in camera. But the digital technology is so good now that you can just play around until it’s right. The great thing about Sam Mendes is he doesn’t have the camera do stupid moves that telegraph the fact that what you are seeing is CG. Everything is shot as if you were really there.”

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